QRM II Graded Assignment (6)

Period 1, 2026

group 6,

10-10-2026

```
movies4 = read.delim("movies4.tsv", header=T)
```

0 Introduction

This assignment is to be completed in groups of 4-5. Further, all students in your group need to be assigned to the same R tutorial group (Friday's tutorial). You can sign yourself up for a group on Canvas. Please do so **before the start of your first R tutorial on Friday September 4th.** You can use the Discussion Board in Canvas if you do not have a group yet or if your group is incomplete.

The assignment has 5 parts, and each part corresponds to the course material of that week (with the exclusion of week 6, for which there is no R programming material).

You are supposed to hand in these assignments on Canvas at the following dates:

- **Deadline 1** *Sunday September 27th, at 23:59pm*: you are supposed to hand in weeks 1, 2, and 3 of this assignment. This will determine 18% of your overall course grade
- **Deadline 2** *Sunday October 11th, at 23:59pm*: you are supposed to hand in weeks 4, and 5 of this assignment. This will determine 12% of your overall course grade

The R tutorials (each Friday) will consist of two halves. During the first half, you will discuss the tutorial exercises. These can be downloaded separately from Canvas. During the second half, you can work on this graded assignment within your own group. The purpose is that you find out how to work with R for doing statistical analyses by yourself. The tutorial exercises are meant to teach you basic commands to get you started, but to answer the problem sets in this assignment, you might need to research your own solutions, and use functions and commands not described in the tutorial exercises. Learning how to solve your own research problems is integral part of learning R. When you and your group get stuck on how to approach an exercise, the hierarchy in finding your way is as follows:

- use the concepts from the tutorial exercises;
- use the cheat sheets available on Canvas;
- use Google, YouTube, StackOverflow, or another website;
- ask the teacher.

The use of generative AI is **not** permitted and may result in a grade of 0. See the AI protocol in the course manual for details.

To answer the assignment, you can simply fill out this R markdown document. There are designated places which you can fill with R code. There are also designated spaces for you to answer each question. Often, the structure of an answer will be as follows. First, you type the R code in the designated box. This will show

how you analyzed the data to get the answer to the question. Below the box for the R code, you will then summarize your answer to the question, i.e. what are the conclusions that you draw from the data analysis?

When handing in, you are supposed to submit this .Rmd file, and a knitted version of this document. You can knit this document to pdf, word, or html. Knitting to pdf requires you to have a .tex distribution installed on your computer. Knitting to Word requires you to have Word installed.

The exercises are designed such that you should be able to finish the majority of them during the tutorial each week. If you are not able to finish them fully during that time, you are expected to work on it in your own time using the computers on campus or your own device. It is best to meet as a group in-person when working together. If you want to work remotely, github is a good platform to guarantee smooth collaboration. Alternatively, you can email this .Rmd file back and forth to one another as a group, but this is not recommended as it is more cumbersome.

We encourage you to keep your code blocks, printing statements, and final answers, as short as possible. In any case, there is a page limit of 6 pages per week, which encompasses the total length of this document which consists of the questions, your coding lines, and your answers. When your answers to questions of the respective week exceed this page limit, they will not be graded, resulting in zero points.

1 Week 1

1. Find the dataset "movies4.tsv" on Canvas. Describe your data set: How many observations does it have. How many variables are there? How many subjects? What consists of a subject? [**4 points**]

```
#WRITE YOUR CODE HERE
movies4 <- read.delim("movies4.tsv", header = TRUE)
nrow(movies4)
```

```
## [1] 808
```

```
ncol(movies4)
```

```
## [1] 19
```

```
#hernoemen row naar subjects en colomn naar variables keer elkaar = observations

subjects <- nrow(movies4)
variables <- ncol(movies4)
observations <- subjects * variables
print(subjects)
```

```
## [1] 808
```

```
print(variables)
```

```
## [1] 19
```

```
print(observations)
```

```
## [1] 15352
```

Your Answer:

The dataset has 15352 observations, 19 variables, 808 subjects. A subject consists of a unit that is being researched. In this case different movies.

2. Which of the following types of variables are present in your data set? (i) nominal; (ii) ordinal; (iii); continuous; (iv) discrete. If present, name one example of such a variable present in your data set. [**4 points**]

```
head(movies4, n=4)
```

```
##   index  budget                                              keywords
## 1  2256 0.0e+00                                                  <NA>
## 2  2473 1.5e+07                barcelona spain menage a trois author
## 3  1514 3.2e+07           gunslinger revenge prairie shootout pistol
## 4  1512 3.2e+07 robbery double life dual identity small town indiana
##   original_language                                    title popularity
## 1                en The Oogieloves in the Big Balloon Adventure    0.28576
## 2                en                     Vicky Cristina Barcelona   32.75825
## 3                en                       The Quick and the Dead   16.47336
## 4                en                        A History of Violence   34.62874
##   release_date  revenue runtime vote_average vote_count  genre release_year
## 1   2012-08-29        0      88          2.2          8 Family         2012
## 2   2008-08-15 96408652      96          6.7       1020  Drama         2008
## 3   1995-02-09 18552460     107          6.3        427 Action         1995
## 4   2005-09-23 60740827      96          6.9        832  Drama         2005
##   release_month release_day        first_actor first_actor_gender
## 1             8          29  Christopher Lloyd               male
## 2             8          15 Scarlett Johansson             female
## 3             2           9       Sharon Stone             female
## 4             9          23    Viggo Mortensen               male
##   director_first_name director_gender
## 1             Matthew            male
## 2               Woody            male
## 3                 Sam            male
## 4               David            male
```

Your Answer:

Genre is nominal, ordinal is not present, popularity is continuous and runtime is discrete.

3. A movie studio wants to know which types of movies give maximal revenue. Perform the following steps to provide the movie studio with an analysis which corresponds to their request:

a. Summarize the variable revenue and report its mean, median, maximum, and minimum. [**2 points**]
b. Which movie has the highest revenue in your data set and how much is this revenue. Which movie has the lowest and how much is its revenue? If multiple movies have the exact same highest or lowest revenue, give only one example. [**2 points**]
c. Create a histogram of the variable revenue. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Describe the shape of the histogram. What does this tell you about the nature of making money in the movies industry? [**2 points**]

d. Add a new variable to your data set: the log of revenue. When creating this variable, what happens to movies for which revenue is zero? What then happens when you calculate the mean of log revenue? **[2 points]**
e. For movies that have a revenue of zero, replace log of revenue with "NA". What is now the mean of log of revenue? Create a histogram for log revenue, again with an appropriate title, x- and y-axis labels. Describe the shape of this histogram. **[2 points]**
f. For the variables revenue and log of revenue calculate their skewness and kurtosis. How do these numbers relate to the shape of the histograms that you estimated under d.) and e.)? **[2 points]**

For each step, you should provide first all the code you used to answer the question and then formulate an answer using full sentences.

Each week consists of 1, 2, or 3 subquestions. The total amount of points you can earn per week is 20 points.

Step a

```
#WRITE YOUR CODE HERE
 movies4 %>%
  summarise(mean_revenue = mean(revenue),
            median_revenue = median(revenue),
            maximum_revenue = max(revenue),
            minimum_revenue = min(revenue))
```

```
##   mean_revenue median_revenue maximum_revenue minimum_revenue
## 1     82727885       14560504      1519557910               0
```

Your Answer:

The mean is 82727885, median is 14560504, maximum is 1519557910 and the minimum is 0. *Step b*

```
max(movies4$revenue)
```

```
## [1] 1519557910
```

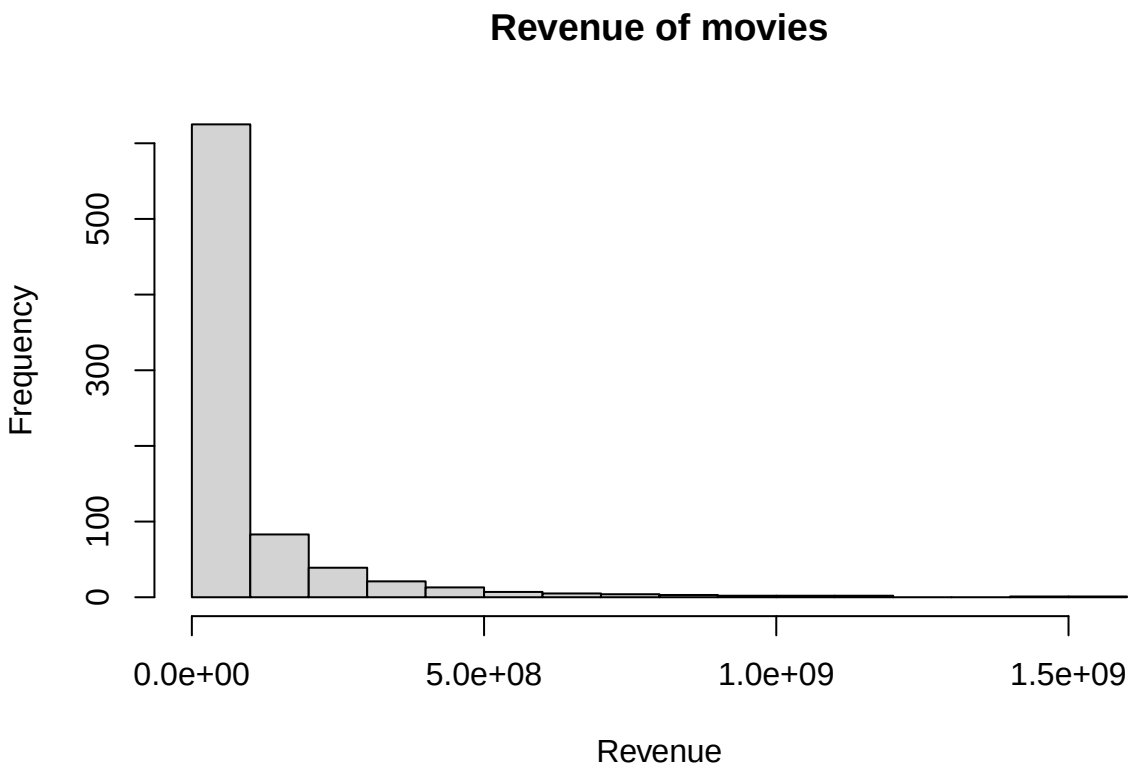
```
min(movies4$revenue)
```

```
## [1] 0
```

Your Answer:

the movie with the highest revenue is The Avengers with a revenue of 1519557910 and an example of a movie with the lowest revenue is Red Dog with a revenue of 0. *Step c*

```
hist(movies4$revenue,
     main = "Revenue of movies",
     xlab = "Revenue",
     ylab = "Frequency")
```



Your Answer:

The histogram is skewed right. This tells us that it is hard to make Revenue from movies in the movie industry. Most movies will not earn a lot. *Step d*

```
movies_LogRevenue <- log(movies4$revenue)
print(movies_LogRevenue)
```

```
##   [1]      -Inf 18.384107 16.736113 17.922127 19.084238      -Inf      -Inf
##   [8]      -Inf      -Inf 10.747100 18.918470 18.637984      -Inf 18.255485
##  [15] 18.490369      -Inf      -Inf      -Inf 18.372660 19.225658      -Inf
##  [22] 20.105195 19.125227 17.196247      -Inf 18.972547      -Inf      -Inf
##  [29] 16.578277      -Inf      -Inf 18.293515 17.801258 17.031271 19.052339
##  [36] 12.706993      -Inf      -Inf      -Inf 18.543078 15.407136      -Inf
##  [43] 19.346038 17.053133      -Inf 18.264990 20.603002 16.119826 14.900016
##  [50]      -Inf 16.090411 15.087134 19.692163 18.877394 19.654616 14.976503
##  [57] 17.825704 14.945238 20.865897 12.945533 10.947732 12.726270 20.590829
##  [64] 18.095755 17.455866 17.067303 14.856785      -Inf 17.078275 19.176213
##  [71] 19.153415      -Inf 18.263396      -Inf 15.908505 15.158411      -Inf
##  [78]      -Inf 17.278518  3.135494 16.523561      -Inf 17.372506 16.395443
##  [85]      -Inf      -Inf      -Inf 18.140928 20.168977      -Inf 17.648850
##  [92]      -Inf 18.462819 17.607644 18.434298      -Inf 13.035131 18.767153
##  [99] 16.873490 16.957907 19.191703 17.051609 16.524105 15.737339 20.592297
## [106] 19.459439 20.291557      -Inf 18.380593 17.750247 18.914626 17.742468
## [113] 19.123569 19.264906 15.175864 18.581842 17.329536 16.569153 17.707331
## [120] 16.692612      -Inf      -Inf 17.238238 15.607868 16.732868      -Inf
```

```
## [127] 15.016992 15.261457 14.220976 19.030099      -Inf      -Inf 14.507561
## [134]      -Inf 16.431574 14.500959      -Inf 19.556308 19.969287 18.543069
## [141] 16.644609      -Inf      -Inf 18.685802      -Inf      -Inf 17.655956
## [148]      -Inf      -Inf 17.588113      -Inf 18.168952 16.161473 18.785324
## [155]      -Inf 19.527489 16.522227 19.013726 18.017033      -Inf 19.426225
## [162] 18.565674 17.180438 13.223471      -Inf      -Inf 17.549930      -Inf
## [169] 19.249558 16.927091 15.131314      -Inf 17.848382 18.458857 18.152985
## [176]      -Inf      -Inf 16.531306 17.148017      -Inf 16.948350 13.558329
## [183]      -Inf 15.672628 11.549595 17.934373 18.538319 17.348534 16.728399
## [190] 20.438534 17.373649 17.669375 15.498246 16.184524      -Inf      -Inf
## [197] 19.146675 19.179268      -Inf 18.836054      -Inf 20.485315      -Inf
## [204]      -Inf      -Inf 19.476755 16.824939 15.894082 14.172695      -Inf
## [211]      -Inf 17.801349 16.656768 17.543521 18.623171 18.151922 16.924578
## [218] 16.786132 17.916960  6.674561 20.334577 14.850869 18.172227 18.902083
## [225] 13.683466 17.642341      -Inf 16.048084      -Inf 17.036464 18.320953
## [232] 19.500642 18.328526 14.205981      -Inf 19.224418      -Inf      -Inf
## [239]      -Inf 18.045687      -Inf 18.938316 17.564718      -Inf      -Inf
## [246] 18.733782 19.356775      -Inf      -Inf 16.939233 19.764932 18.994959
## [253]      -Inf 17.072320      -Inf 18.894322 19.357847      -Inf      -Inf
## [260]      -Inf 16.111081 16.776762      -Inf  2.564949 17.486735      -Inf
## [267] 18.769750      -Inf 12.142462 19.189232 16.778284 17.534606      -Inf
## [274] 17.833402 18.386283 19.941717 18.381825 11.461632 17.796825      -Inf
## [281]      -Inf 16.884527 18.815503 18.048590 18.891175 15.171086      -Inf
## [288] 18.440483 20.013722 14.865182 18.138252 16.784802      -Inf 19.238243
## [295] 10.324531 16.647553 17.063518 15.798745 19.202814      -Inf 15.684570
## [302] 18.464491 14.086051      -Inf 16.182749 17.284398 19.433348 15.988479
## [309]      -Inf 17.601768 18.236286      -Inf 17.916145 13.649106 18.528106
## [316]      -Inf      -Inf      -Inf 18.471382 18.545636      -Inf 17.327857
## [323]      -Inf      -Inf 18.130163 19.725713 18.760467 18.127766      -Inf
## [330]      -Inf 16.692062 15.333304 18.554171      -Inf      -Inf 17.568478
## [337] 17.900118 15.740117 15.761421      -Inf 16.199563 16.723348 14.939407
## [344]      -Inf 20.198577 21.141685      -Inf 19.665512      -Inf 18.908277
## [351] 15.219111 19.998441 17.118263      -Inf      -Inf      -Inf 17.358224
## [358] 18.303284      -Inf 19.042155 17.281410 18.425586 18.827945 18.097957
## [365] 16.972833 19.482103 16.898308      -Inf 14.102366 18.561326 17.113339
## [372] 16.421984      -Inf      -Inf 17.267238 18.396093 17.984140      -Inf
## [379] 18.237747 17.822901 17.354058      -Inf 16.131012 18.919042 17.778189
## [386] 19.752736      -Inf 19.172149      -Inf      -Inf 20.326968 16.140770
## [393] 17.612071 16.093814 17.434817      -Inf 15.776165 16.486897 19.718422
## [400]      -Inf 13.816947 17.928234 17.536284 17.133720 19.259871 19.341648
## [407]      -Inf      -Inf 20.420267      -Inf 17.471278 15.685964      -Inf
## [414]      -Inf 18.143367 17.978394 20.683485 18.087513 14.711944 18.177063
## [421] 17.335621 19.945344      -Inf      -Inf 16.597653 17.599700      -Inf
## [428]      -Inf      -Inf 17.040280 15.201805  4.521789 17.963826 17.515643
## [435] 18.752518 11.250274 16.352321 17.374703 12.898247 18.819609 17.917730
## [442] 18.368711      -Inf      -Inf      -Inf 16.185400 14.480078      -Inf
## [449] 16.428832      -Inf      -Inf 16.828629 17.687294      -Inf      -Inf
## [456] 19.629347 19.005014      -Inf 18.969072      -Inf 15.087018      -Inf
## [463] 17.952542 15.725778 19.408088      -Inf      -Inf 14.998337 14.638543
## [470] 19.158784      -Inf 13.343149 18.139721 16.698581      -Inf 12.310433
## [477] 18.831576      -Inf 20.132313 16.817093 19.697092      -Inf      -Inf
## [484] 20.341024      -Inf 17.763960 15.434640      -Inf 16.188109 16.686525
## [491] 18.974333 15.211206      -Inf 16.554982      -Inf 16.868122 16.367126
## [498]      -Inf 17.771039      -Inf 16.896812 17.555409 14.862922      -Inf
```

```
## [505] 14.583088 18.577684 18.574885      -Inf      -Inf      -Inf      -Inf
## [512]      -Inf      -Inf 16.886869 17.466230 15.503670 17.163716      -Inf
## [519] 19.941594      -Inf      -Inf 19.054299 15.285864 12.959280 18.575687
## [526] 15.962027      -Inf      -Inf 17.134872      -Inf      -Inf 14.847931
## [533] 13.367660      -Inf      -Inf 19.568580 13.083230 17.227686      -Inf
## [540] 17.194158 14.697405 18.681755 20.727814 17.453097 16.005673 15.202085
## [547] 17.509453 19.861961      -Inf 20.114340 18.200009 18.874220 10.879669
## [554]      -Inf 19.555111      -Inf 18.427201      -Inf 19.092277 19.187517
## [561] 19.204932 19.028334      -Inf      -Inf      -Inf      -Inf 16.511600
## [568]      -Inf 18.934830      -Inf      -Inf 16.500702 14.862528 16.684899
## [575]      -Inf 16.181305      -Inf 17.716875 13.180723      -Inf 18.755754
## [582] 16.535443      -Inf 18.630195 16.482739 19.314839      -Inf 18.180343
## [589] 16.528317      -Inf 18.606795 20.432730 13.997832      -Inf 19.123908
## [596]      -Inf 19.679466 18.168526 16.712762 18.577519 16.908564      -Inf
## [603]      -Inf 17.281803 20.826329 17.561084      -Inf 19.685508 19.011919
## [610] 18.950350 15.114358 13.302424 18.602509 19.903656 19.479802      -Inf
## [617]  9.768355 18.594637 16.035417 15.761744      -Inf      -Inf 19.953238
## [624]      -Inf 16.769588 17.134973 19.367889 17.517983 11.243502      -Inf
## [631] 15.584971 17.153756 18.821274      -Inf 21.063590 16.611225 17.469526
## [638] 15.802312 16.866026 17.554271      -Inf      -Inf 18.227027      -Inf
## [645] 16.552935 17.606205 12.597270 17.127435      -Inf 17.648949 17.599700
## [652]      -Inf      -Inf 18.627037 16.837359      -Inf 17.846379 15.929235
## [659] 18.868601 16.795819 18.376697      -Inf 16.995562 20.788088      -Inf
## [666] 18.343178      -Inf 18.118984      -Inf 14.877213 20.134456 16.781453
## [673]      -Inf 18.858534 15.560914 11.695247 19.689578      -Inf 15.272255
## [680] 15.561419 20.227356 17.575106 15.950796 17.602130      -Inf      -Inf
## [687]      -Inf 18.420681 20.699460 16.678132 18.664931 16.831857      -Inf
## [694] 16.287856      -Inf 14.768252 16.419816      -Inf      -Inf 17.937777
## [701] 18.055561 16.644650 14.418024 19.142194 19.138519      -Inf 18.242981
## [708] 15.240948      -Inf 17.575092 18.909053 18.676732 19.975086 16.204626
## [715] 17.301219 18.318200 16.173664 15.842011 18.366009      -Inf 16.229321
## [722]      -Inf      -Inf 19.725958      -Inf 17.785802      -Inf      -Inf
## [729] 19.905204      -Inf      -Inf 13.242288 17.632764 15.713643      -Inf
## [736] 17.506887 19.201567      -Inf      -Inf 17.748390 18.259712      -Inf
## [743]      -Inf 18.550459 14.028697      -Inf 17.889581 18.971103 19.630103
## [750] 15.976511 15.051811 18.038881 19.825804      -Inf 12.746479 20.040069
## [757] 18.324290 16.547856 17.035901      -Inf      -Inf 18.777082 18.443971
## [764] 16.317440 16.038779      -Inf 19.193035 13.258308 17.970983      -Inf
## [771]      -Inf 18.739510      -Inf 16.926084 13.311235 19.730475      -Inf
## [778] 19.764217 18.136003      -Inf      -Inf      -Inf 19.273863 16.887634
## [785] 14.631190      -Inf 18.839630      -Inf 17.483451      -Inf 17.048092
## [792]      -Inf      -Inf 17.846178 18.227905      -Inf 19.666086 14.253091
## [799] 19.868466      -Inf 17.750130 18.833580 15.341794 15.016779 16.641206
## [806]      -Inf 15.734606 18.735366
```

```
mean(movies_LogRevenue)
```

```
## [1] -Inf
```

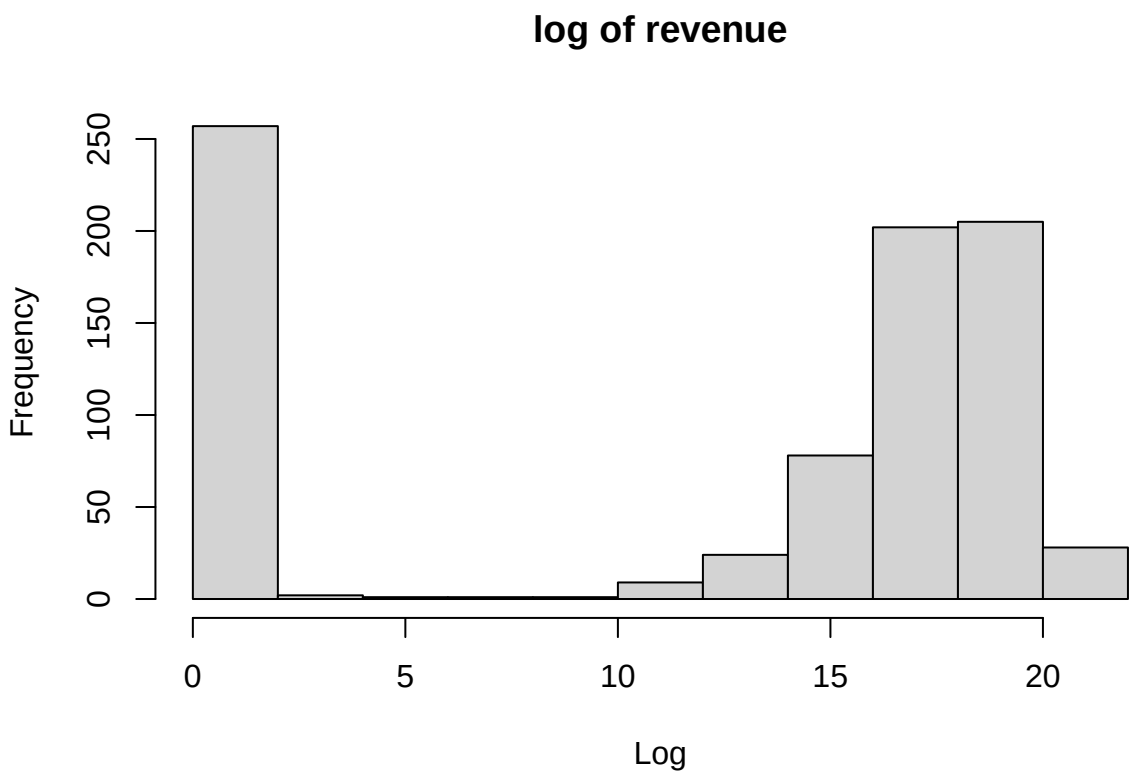
Your Answer:

movies with no revenue will show -inf since the log of 0 does not exist. The mean also shows -inf since the values of -inf are still included when calculating the mean. *Step e*

```r
movies_LogRevenueNA <- log(movies4$revenue+1)
movies4$revenue[movies_LogRevenueNA == 0] <- NA
mean(movies_LogRevenueNA, na.rm = TRUE)
```

```
## [1] 11.78363
```

```r
hist(movies_LogRevenueNA,
     main = "log of revenue",
     xlab = "Log",
     ylab = "Frequency")
```



Your Answer:

The mean is 17.2798 and the shape of the histogram is left skewed.

Step f

```r
skewness(movies4$revenue, na.rm = TRUE)
```

```
## [1] 3.261763
```

```r
kurtosis(movies4$revenue, na.rm = TRUE)
```

```
## [1] 16.35398
```

```r
skewness(movies_LogRevenueNA, na.rm = TRUE)
```

```
## [1] -0.6475635
```

```r
kurtosis(movies_LogRevenueNA, na.rm = TRUE)
```

```
## [1] 1.554936
```

Your Answer:

The skewness is 3.261763 and the kurtosis is 16.35398 for the variable revenue. The skewness is above 0 which explains that it is skewed right. The kurtosis of 16 explains why the histogram has high frequency to the left side and is not flat. The skewness for the log of revenue is -1.946866 which tells us that it is skewed left. The kurtosis of 10.87042 tells us that the that the histogram is not flat.

2 Week 2

1 Is your dataset movies4.tsv the full population, or is it a sample of a larger population? If the latter, how would you describe the full population? [**4 points**]

Your answer here:

The dataset is a sample of a larger population since the index is missing a lot of numbers. We would describe the full population as all movies from 1990 to 2016.

2 For this question, you will assume that your data set is the full population.

a. What is the mean of the variable vote_average in your data set? [**2 points**]
b. Create a new data set, called movies_sample. Make sure that it is a random sample of your data set of 25 movies. What is the mean of vote_average in this random sample? How does it compare to the mean of vote_average in 2a? [**2 points**]
c. In a for loop, create 100 different samples of 25 movies, as in b, and estimate the mean within each sample. Save the mean of each sample in a vector called sample_means. So the first position of the vector would have the mean of the first sample, the second position the mean of the second sample, etc. Print the start of this vector. [**2 points**]
d. Summarize and make a histogram of sample_means. What is the mean, standard deviation and shape of its distribution? [**2 points**]
e. Using the formulas of the *Central Limit Theorem*, give the distribution of the sample mean of vote_average in a sample of 25 movies from your population (including its expected value and standard deviation). How do these values compare to your answers at d.)? [**2 points**]

Step a

```r
#WRITE YOUR CODE HERE
mean(movies4$vote_average)
```

```
## [1] 6.039356
```

Your Answer:

The mean is 6.039356 *Step b*

```r
#WRITE YOUR CODE HERE
movies_sample <- rnorm(25, 6.039356, sd(movies4$vote_average))
mean(movies_sample)
```

```
## [1] 6.072237
```

Your Answer: The mean of the sample set does not match the average of the total movies, because the sample set only includes 25 random movies, and thus is a bit skewed because they are not representative of all the movies.

Step c

```r
#WRITE YOUR CODE HERE
sample_means <- c()
  for(i in c(1:100)){
    sample_means[i] <- mean(rnorm(25, 6.039356, sd(movies4$vote_average)))}
    head(sample_means)
```

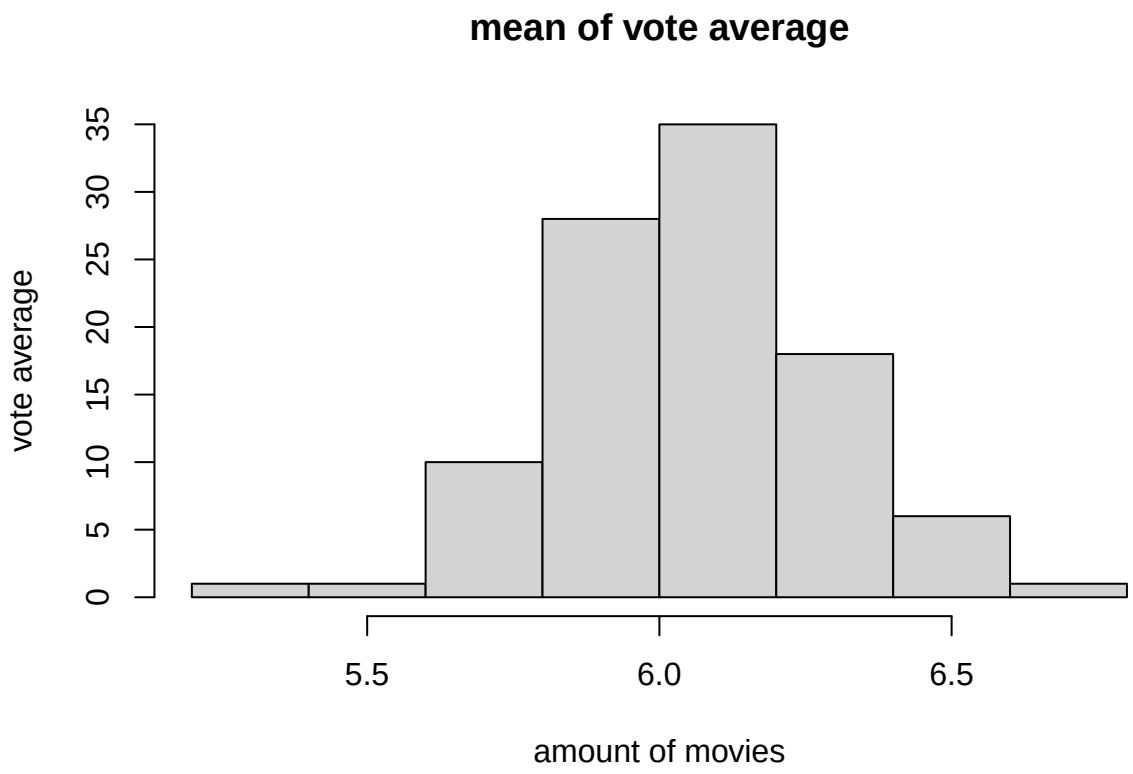
```
## [1] 6.084108 5.670273 6.296584 6.053584 5.719045 5.873233
```

Your Answer:

5.984406 6.219849 6.091184 6.334625 6.456935 6.357936

Step d

```r
#WRITE YOUR CODE HERE
hist(sample_means,
     main = "mean of vote average",
     xlab = "amount of movies",
     ylab = "vote average")
```




```
mean(sample_means)
```

```
## [1] 6.054075
```

```
sd(sample_means)
```

```
## [1] 0.2390495
```

Your Answer:

The mean is 6.020662 and the standard deviation is 0.2515497. The shape of the histogram is skewed left.

Step e

Your Answer:

```
m <- mean(movies4$vote_average)
sd <- sd(movies4$vote_average)/sqrt(25)
print(m)
```

```
## [1] 6.039356
```

```r
print(sd)
```

```
## [1] 0.2369204
```

We expected a low standard deviation, because of the normal distribution. This is the case because the standard deviation is 0.2515497 so this tells us that the graph in d is indeed normally distributed. The expected value is aswell very close to the middle of the graph also is expected in such a normally distributed graph.

3. Is there a variable in your data set which you would expect to follow a discrete uniform distribution? Explain, without looking at the data, why you expect this data to be discrete uniform. If the data would indeed follow a discrete uniform distribution, what would you expect the mean and standard deviation of this variable to be? Next, plot a histogram of the variable, and plot an appropriate discrete uniform distribution on top of this histogram. Comment whether the variable indeed follows this expected distribution. If the distribution deviates from the discrete uniform distribution, explain why this is the case. [**6 points**]

Your Answer:

I would expect the release month of a movie to follow a discrete uniform distribution because there are 12 possible months, and without looking at the data, each month could be expected to be equally likely for a movie to be released. Therefore, each month would have a probability of 1/12.

For a discrete uniform distribution from 1 to 12, the expected mean is 6.5 and the expected standard deviation is approximately 3.45. we chose for the release month variable because its logical that the whole year movies were released because there is not a particular month where the benefits are larger to release a new movie. Thats why we expect that release months does not have that much impact.

```r
#WRITE YOUR CODE HERE

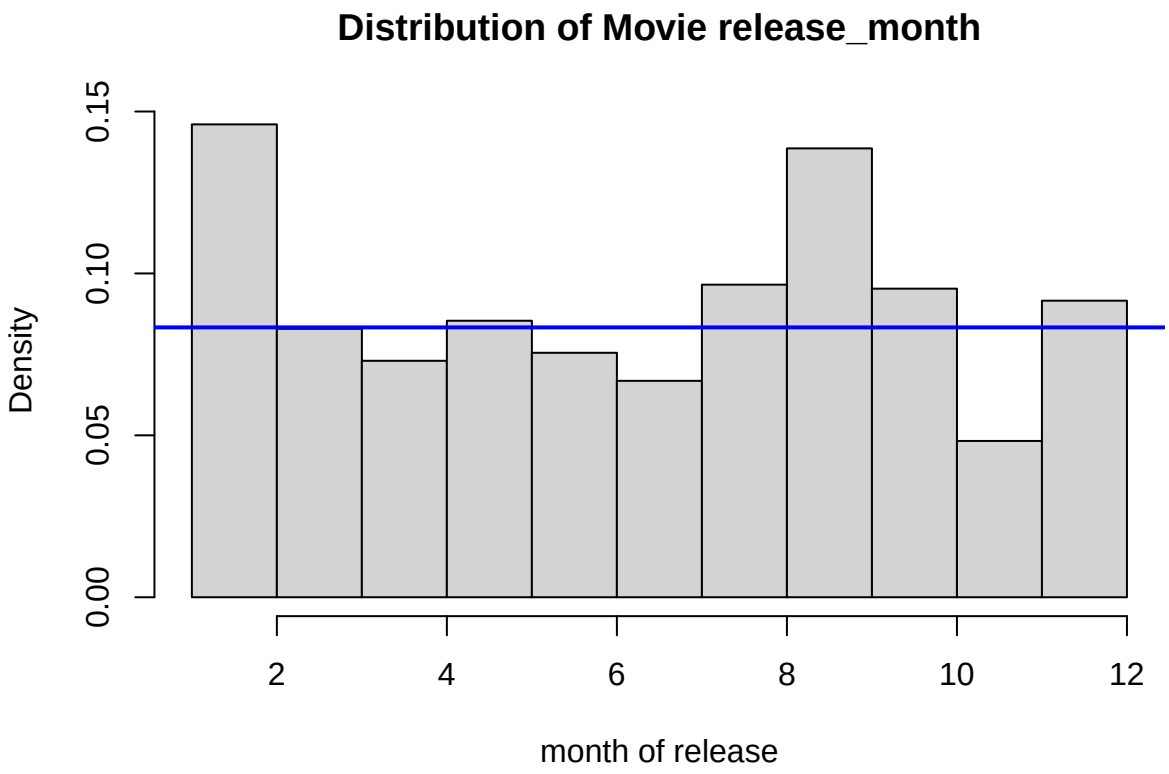
summary(movies4$release_month)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.000   4.000   7.000   6.708   9.000  12.000
```

```r
sd(movies4$release_month)
```

```
## [1] 3.373684
```

```r
hist(movies4$release_month,freq=FALSE,
     main="Distribution of Movie release_month",
     xlab="month of release")
abline(h=1/12,col="blue",lwd=2)
```



yes the distrubution is similiar as the discrete uniform distrubution what we would expect.

Your Answer:

Write your formulated response here. There are 808 movies in the dataset, so if the distribution were perfectly uniform, we would expect 67.3 movies to be released in each month. The observed frequencies range from 39 movies in November to 112 movies in September.

The histogram therefore does not follow a perfectly discrete uniform distribution. September has considerably more movie releases than expected, while November has considerably fewer. The deviation from the discrete uniform distribution can be explained by the fact that movie releases are not randomly distributed throughout the year. Movie studios may choose particular release months based on factors such as holidays, audience availability or competition. Therefore, some months can have more movie releases than others.

3 Week 3

For the next part of the assignment, assume that the movies in your data frame are a random sample of a larger population of movies.

1. Give a 90 percent confidence interval for the variable *revenue*. Interpret the result. [**2 points**]

```
#WRITE YOUR CODE HERE


x <- movies4$revenue

mean(x, na.rm = TRUE)
```

```
## [1] 121314213
```

```
sd(x, na.rm = TRUE)
```

```
## [1] 197480675
```

```
sum(!is.na(x))
```

```
## [1] 551
```

```
t.test(x, conf.level = 0.90)
```

```
##
##  One Sample t-test
##
## data:  x
## t = 14.42, df = 550, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 90 percent confidence interval:
##  107452778 135175648
## sample estimates:
## mean of x
## 121314213
```

Your Answer:

The 90% confidence interval for the mean revenue is [107,452,778, 135,175,648].

Assuming that the movies in our data frame are a random sample from a larger population of movies, we are 90% confident that the true mean revenue of the population of movies is between $107,452,778 and $135,175,648.

2

a. How many movies in your sample were directed by a male director? How many by a female director? [**2 points**]
b. Test the null hypothesis that the chance that a movie is being directed by a male or female director is equal. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Use an appropriate R function to test your hypothesis, and state your conclusion. Also give the p-value that corresponds to your test statistic. [**2 points**]
c. Recode revenue such that it is measured in millions of dollars. What is the variance of revenue for movies with a male director? What is the variance of revenue for movies with a female director? [**2 points**]
d. Assume that the variance of revenue for movies with a male director in your data is the same as the variance for movies with a male director in your population (that is, there is no sampling uncertainty in this estimate). Now, set up a hypothesis test to test for the null hypothesis that the variance of revenue for movies with a female director is equal to the variance for movies with a male director. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Calculate your test statistic and tell how your test statistic is distributed, and state your conclusion. Also give the p-value that corresponds to your test statistic. [**2 points**]

step a

```r
#WRITE YOUR CODE HERE
sum(movies4$director_gender == "male", na.rm = TRUE)
```

```
## [1] 685
```

```r
sum(movies4$director_gender == "female", na.rm = TRUE)
```

```
## [1] 80
```

Your Answer:

In the sample, 685 movies were directed by a male director, while 80 movies were directed by a female director.

step b

```r
#WRITE YOUR CODE HERE
binom.test(685, 765, p = 0.5, alternative = "two.sided")
```

```
##
##  Exact binomial test
##
## data:  685 and 765
## number of successes = 685, number of trials = 765, p-value < 2.2e-16
## alternative hypothesis: true probability of success is not equal to 0.5
## 95 percent confidence interval:
##  0.8715412 0.9162048
## sample estimates:
## probability of success
##              0.8954248
```

Your Answer:

The null and alternative hypotheses are:

H0:p=0.5, meaning that the probability of a movie being directed by a male or female director is equal.

HA:p=/0.5, meaning that the probabilities are not equal.

I use a significance level of a=0.05.

Using a two-sided binomial test in R with binom.test(685, 765, p = 0.5), the p-value is extremely small ($p < 0.001$).

Since the p-value is smaller than the significance level of 0.05, we reject the null hypothesis. There is strong statistical evidence that the probability of a movie being directed by a male director is not equal to the probability of it being directed by a female director.

step c

```r
#WRITE YOUR CODE HERE

movies4$revenue_millions <- movies4$revenue / 1000000


var(movies4$revenue_millions[movies4$director_gender == "male"],
    na.rm = TRUE)
```

```
## [1] 39453.26
```

```
var(movies4$revenue_millions[movies4$director_gender == "female"],
    na.rm = TRUE)
```

```
## [1] 21607.04
```

Your Answer:

Revenue was recoded from dollars to millions of dollars by dividing revenue by 1,000,000.

The variance of revenue for movies with a male director is 39,453.26 (million dollars)².

The variance of revenue for movies with a female director is 21,607.04 (million dollars)².

step d

```
#WRITE YOUR CODE HERE

F <- 21607.04 / 39453.26
F
```

```
## [1] 0.5476617
```

```
p_value <- 2 * min(
  pf(F, df1 = 79, df2 = 79),
  1 - pf(F, df1 = 79, df2 = 79)
)

p_value
```

```
## [1] 0.008105479
```

Your Answer:

The null hypothesis is

H0: standard deviation F2=39,453.26

meaning that the population variance of revenue for movies with a female director is equal to the population variance for movies with a male director.

The alternative hypothesis is

HA: standard deviation F2=/39,453.26.

I use a significance level of a=0.05.

The test statistic is

F=21,607.04 /39,453.26=0.5477.

Under the null hypothesis, the test statistic follows an F-distribution with 79 degrees of freedom. The two-sided p-value is 0.0081.

Since the p-value (0.0081) is smaller than the significance level of 0.05, we reject the null hypothesis. There is sufficient evidence to conclude that the variance of revenue for movies with female directors is different from the variance of revenue for movies with male directors.

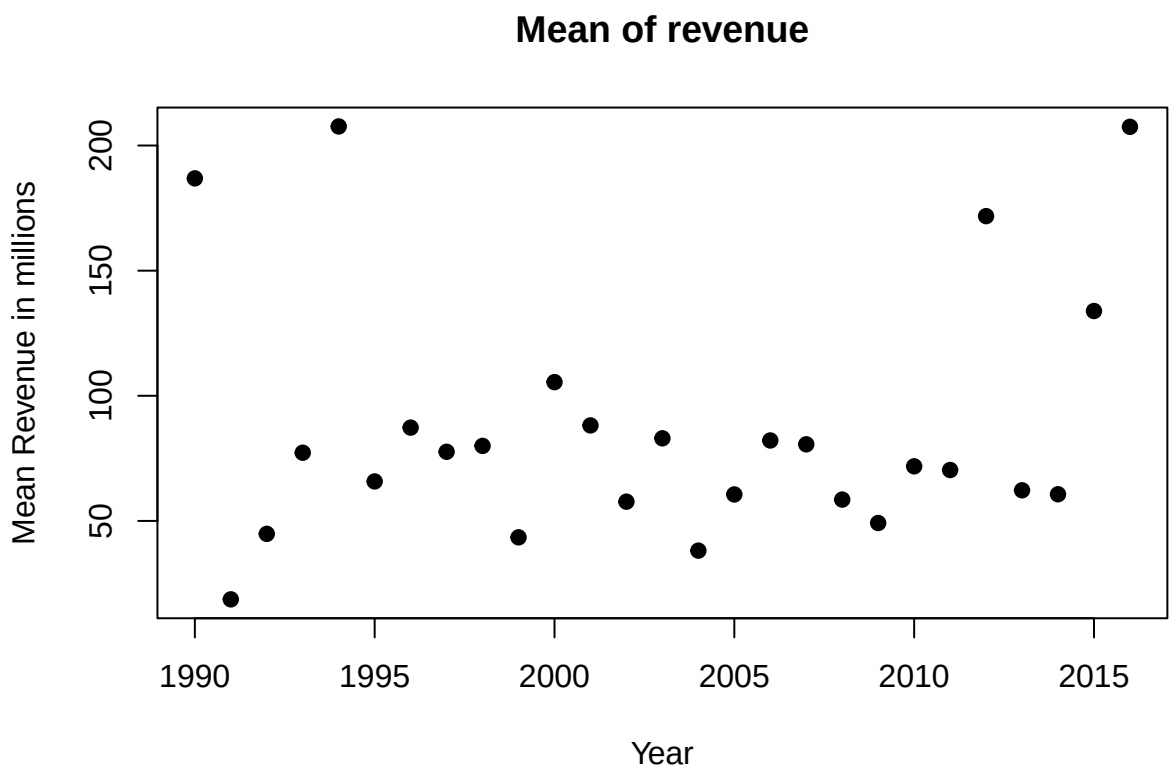
3

a. Create a scatter plot with year on the x axis and mean of revenue within that year on the y-axis. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Is revenue of movies increasing or decreasing over time? **[3 points]**
b. Recreate the scatter plot with year on the x axis and mean_revenue on the y-axis, but now add bars around each point, indicating the 95% confidence interval. **[3 points]**
c. If you look closely at the resulting plot in b.), you will probably see that the confidence intervals in some years are wider than in other years. What are some possible reasons that a confidence interval in a given year can be wide? Can you verify this using your data? **[4 points]**

step a

```
#WRITE YOUR CODE HERE
movies <- read.delim("movies4.tsv", header = TRUE)
mean_revenue <- aggregate(revenue ~ release_year,
                          data = movies,
                          FUN = mean,
                          na.rm = TRUE)

plot(mean_revenue$release_year, mean_revenue$revenue/1000000,
     main = "Mean of revenue",
     xlab = "Year",
     ylab = "Mean Revenue in millions",
     pch = 19)
```



Your Answer:

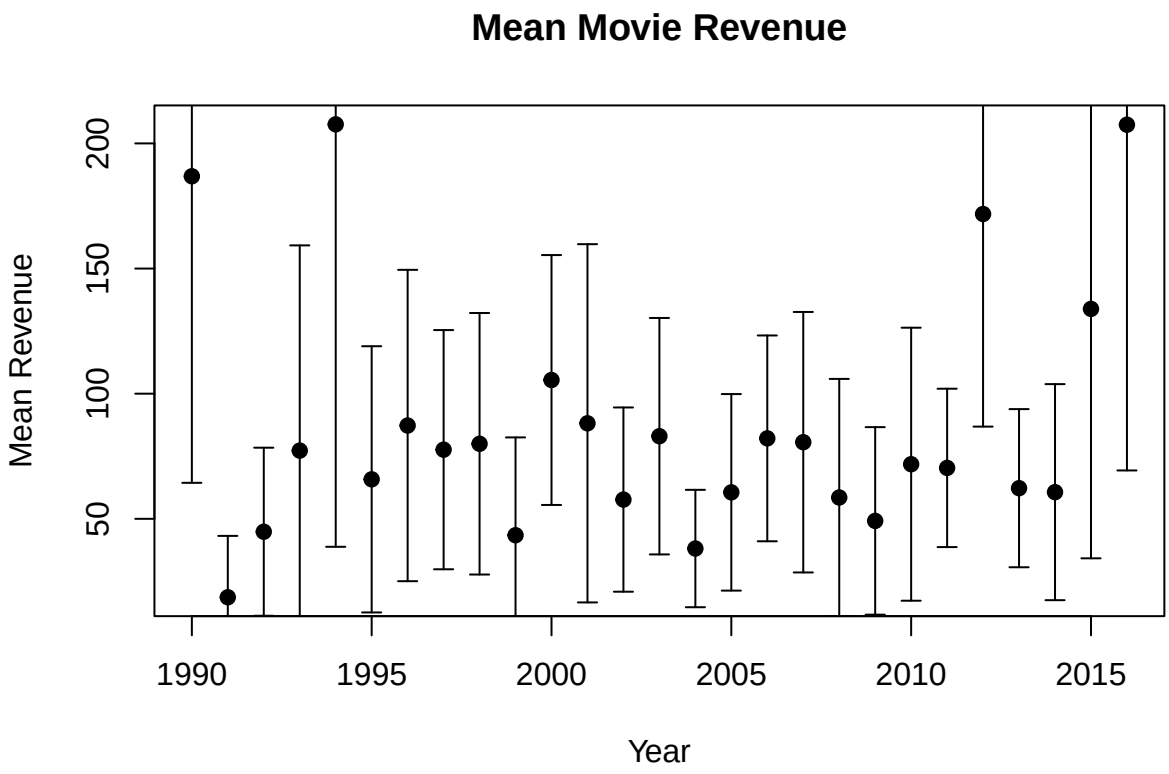
The overall trend is that revenue of movies is decreasing over time.

step b

```r
mean_revenue <- movies %>%
  group_by(release_year) %>%
  summarise(
    mean_revenue = mean(revenue, na.rm = TRUE),
    sd_revenue = sd(revenue, na.rm = TRUE),
    n = sum(!is.na(revenue))
  ) %>%
  mutate(
    se = sd_revenue / sqrt(n),
    lower = mean_revenue - 1.96 * se,
    upper = mean_revenue + 1.96 * se
  )

plot(mean_revenue$release_year,
     mean_revenue$mean_revenue / 1000000,
     main = "Mean Movie Revenue",
     xlab = "Year",
     ylab = "Mean Revenue",
     pch = 19)

arrows(mean_revenue$release_year,
       mean_revenue$lower / 1000000,
       mean_revenue$release_year,
       mean_revenue$upper / 1000000,
       angle = 90,
       code = 3,
       length = 0.05)
```



Your Answer:

The reason for this is that the standard deviations from year to year vary, the higher the standard deviation is the wider the confidence interval needs to be. This is most clearly seen in years with a high standard deviation or a lower N. The N is relevant, because the higher the number of observations you have the narrower the confidence interval is, because you have more data to trust and thus have greater certainty.

step c

```
results <- data.frame(
  year = mean_revenue$release_year,
  sd = mean_revenue$sd_revenue,
  n = mean_revenue$n,
  CI_width = mean_revenue$upper - mean_revenue$lower
)
highest <- results[order(results$CI_width, decreasing = TRUE), ][1:2, ]

lowest <- results[order(results$CI_width), ][1:2, ]

highest
```

```
##    year        sd  n  CI_width
## 5  1994 285619241 11 337580371
## 27 2016 315205264 20 276289595
```

```
lowest
```

```
##    year       sd  n CI_width
## 15 2004 72791859 37 46910279
## 2  1991 30641357  6 49036385
```

4 Week 4

1. There is an argument going on in the movie studio. *Bob* claims that production budgets are getting out of hand, and that the studio should focus on making cheaper movies. *Chantal* disagrees. She tells Bob that "Every dollar we spend on movie production is more than offset by the increase in movie revenue''.

   a. Set up a regression model to test Chantal's claim, and estimate it. That is, estimate:

   $$\text{Revenue}_i = \beta_0 + \beta_1 \text{Budget}_i + \varepsilon_i.$$

   Print a summary of your estimated model. **[2 points]**
   b. What is the estimated value of β_1 and how do you interpet it? **[2 points]**
   c. Test for the null hypothesis that $\beta_1 \leq 1$. Report the p-value and state your conclusion. **[2 points]**
   d. Next, estimate the model

   $$\text{Log Revenue}_i = \beta_0 + \beta_1 \text{Log Budget}_i + \varepsilon_i.$$

   When creating the variables Log Revenue and Log Budget, make sure that movies with a Revenue or Budget of zero are assigned the value "NA". Print a summary of your estimated model. **[2 points]**
   e. What is the estimated value of β_1 and how do you interpet it? **[2 points]**
   f. Which model has better fit? The level-level model or the log-log model? Explain. **[2 points]**
   g. Who do you think is correct? Bob or Chantal? What would you advise the movie studio to do? **[2 points]**

step a

```
#WRITE YOUR CODE HERE
model <- lm(movies4$revenue ~ movies4$budget)
summary(model)
```

```
##
## Call:
## lm(formula = movies4$revenue ~ movies4$budget)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -447658083  -57507248   -6170960   23601366  814123016
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -1.352e+07  7.358e+06  -1.837   0.0667 .
## movies4$budget  3.268e+00  1.192e-01  27.410   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 128400000 on 549 degrees of freedom
##   (257 observations deleted due to missingness)
## Multiple R-squared:  0.5778, Adjusted R-squared:  0.577
## F-statistic: 751.3 on 1 and 549 DF,  p-value: < 2.2e-16
```

Your Answer:

Write your formulated response here.

step b

```
#WRITE YOUR CODE HERE
beta1 <- coef(model)["movies4$budget"]
beta1
```

```
## movies4$budget
##       3.267965
```

Your Answer:

This result means that if the budget increases by one unit, revenue increases by 3.268 units.

step c

```
#WRITE YOUR CODE HERE
model <- lm(movies4$revenue ~ movies4$budget)

beta1 <- coef(model)[2]
se <- summary(model)$coefficients[2,2]

t <- (beta1 - 1) / se

p_value <- pt(t, df.residual(model), lower.tail = FALSE)

p_value
```

```
## movies4$budget
##   1.182461e-62
```

Your Answer:

This p-value is smaller than just about every existing a-value, from which you can conclude that H0 can be rejected and that beta1 is greater than 1

step d

```
#WRITE YOUR CODE HERE
movies4$log_revenue <- ifelse(movies4$revenue == 0, NA, log(movies4$revenue))
movies4$log_budget <- ifelse(movies4$budget == 0, NA, log(movies4$budget))

log_model <- lm(log_revenue ~ log_budget, data = movies4)

summary(log_model)
```

```
##
## Call:
## lm(formula = log_revenue ~ log_budget, data = movies4)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.6073  -0.6971   0.2294   0.8698   3.8941
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.04588    0.77932  -0.059    0.953
## log_budget   1.02996    0.04573  22.523   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.495 on 517 degrees of freedom
##   (289 observations deleted due to missingness)
## Multiple R-squared:  0.4953, Adjusted R-squared:  0.4943
## F-statistic: 507.3 on 1 and 517 DF,  p-value: < 2.2e-16
```

Your Answer:

The estimated regression model is, Log Rêvenue = -0.04588 + 1.02996 Log Budget.

step e

```
#WRITE YOUR CODE HERE
coef(log_model)[2]
```

```
## log_budget
##   1.029956
```

Your Answer:

a 1% increase in budget is associated with a 1.03% increase in revenue in this example.

step f

```
#WRITE YOUR CODE HERE
R_squared_model <- summary(model)$r.squared
R_squared_log_model <- summary(log_model)$r.squared

if (R_squared_model>R_squared_log_model){
  "The level-level model has a better fit"
} else {
  "The log-log model has a better fit"
}
```

```
## [1] "The level-level model has a better fit"
```

Your Answer: Write your formulated response here. The level-level model has a better fit, because the R-squared of the level-level model is greater than the R-squared of the log-log model.

step g

```r
#WRITE YOUR CODE HERE
if (p_value < 0.05 && beta1 > 1) {
  cat("Conclusion: The data support Chantal's claim.\n")
} else {
  cat("Conclusion: The data do not provide sufficient evidence for Chantal's claim.\n")
}
```

```
## Conclusion: The data support Chantal's claim.
```

Your Answer: Write your formulated response here. I think Chantal is correct. If you were to increase the budget, the revenue will be increased even more. The statement made by Chantal is correct. My advice is that the studio should consider the estimated relationship between budget and revenue and choose production budgets carefully. The evidence rejects the idea that higher production spending is automatically wasteful.

2

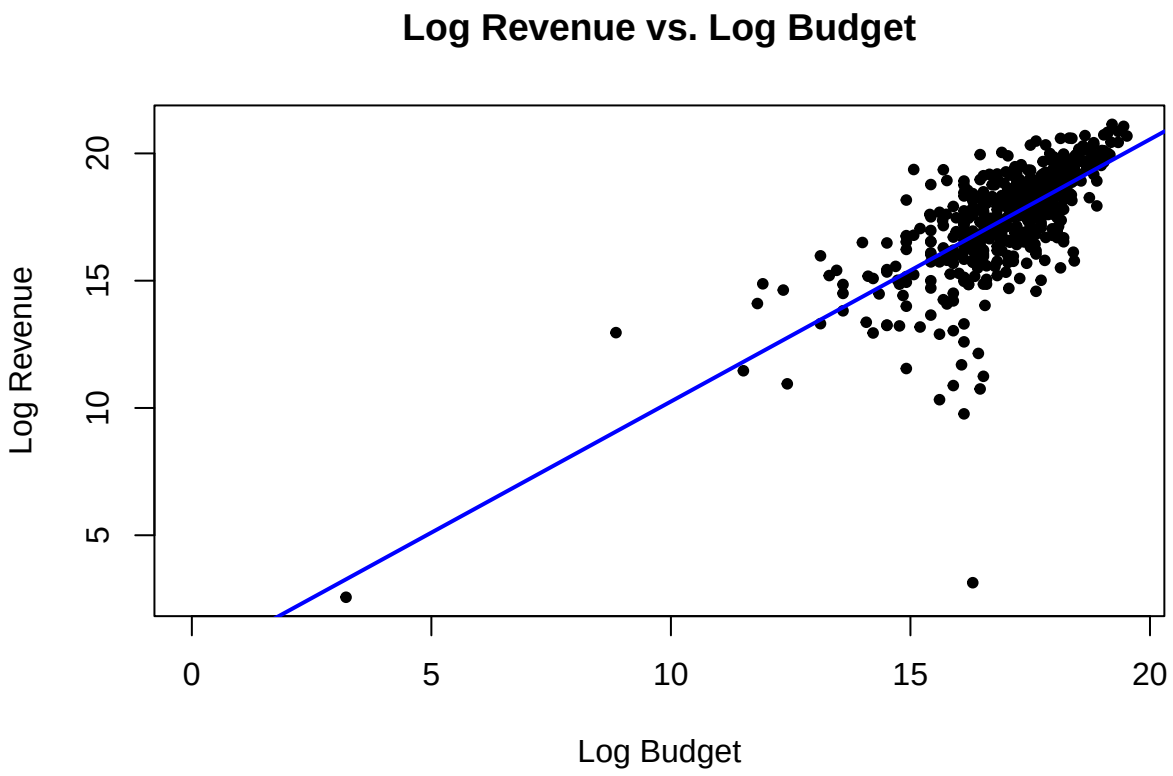
a. Make a scatterplot with log revenue on the y-axis, and log budget on the x-axis. Within the same scatterplot, draw a straight line which summarizes the association between the log of revenue and log of budget. **[3 points]**
b. Using the insights from the scatterplot, your linear model, and your data, find an example of a movie that vastly overperformed. That means, find a movie that had much more revenue than expected, given its budget. For this movie, report its log of revenue, its predicted log of revenue, and its residual. **[3 points]**

step a

```r
#WRITE YOUR CODE HERE
plot(movies4$log_budget, movies4$log_revenue,
     xlab = "Log Budget",
     ylab = "Log Revenue",
     main = "Log Revenue vs. Log Budget",
     pch = 20)

log_model <- lm(log_revenue ~ log_budget, data = movies4)

abline(log_model, col = "blue", lwd = 2)
```



Your Answer:

Write your formulated response here. The scatterplot shows a positive assocation between log revenue and log budget. The fitted regression line indicates that movies with higher log budgets tend to have higher log revenues.

step b

```
#WRITE YOUR CODE HERE

i <- which.max(residuals(log_model))

model_data <- model.frame(log_model)

original_row <- as.numeric(rownames(model_data)[i])

movies4[original_row, c("title")]
```

```
## [1] "The Full Monty"
```

```
c(
  "Log Revenue" = model_data$log_revenue[i],
  "Predicted Log Revenue" = predict(log_model)[i],
  "Residual" = residuals(log_model)[i]
)
```

```
##               Log Revenue Predicted Log Revenue.627              Residual.627
##                 19.367889                 15.473776                  3.894113
```

Your Answer:

Write your formulated response here. The Full Monty is an example of a movie that vastly overperformed. Its actual log revenue is 19.368, while its predicted log revenue was 15.474. The residual is 3.894, this means that the movie's actual log revenue is way higher than what the model predicted given its budget.

5 Week 5

a. Make a frequency table of the variable genre. Which are the three genres that occur the most? [**2 points**]
b. Create a new variable called genre2. Make sure that the three most popular genres remain the same, but that all the other genres get categorized into one genre called "Other". Report a frequency table of this new variable genre2. [**2 points**]
c. Estimate an OLS model which has as dependent variable the log of revenue of a movie, and as independent variable the log of budget, a dummy for each level that genre2 can take, and the year of release. Show a summary of the resulting model and interpret each coefficient. [**4 points**]

The movie studio that you work at is releasing a new movie in 2026. It will be a Drama movie with a budget of 20,000,0000. It will have a runtime of 120 minutes.

d. Estimate a model that is able to predict the revenue of this movie. Give its predicted revenue and include a 90% prediction interval. [**5 points**]
e. Plot the residuals against various independent variables in the model. Do you find any evidence of heteroskedasticity? [**2 points**]
f. One executive at the studio wants to time the release of the movie to a specific month of the year such that they can maximize revenue. Another executive is of the opinion that the month in which the movie is released will not have a big impact on its revenue. Estimate a model that can test whether the month of release impacts the revenue of a movie. Use an appropriate hypothesis test to test whether month of release matters. What would you advise the movie studio regarding the timing of the release of the movie? [**5 points**]

step a

```
#WRITE YOUR CODE HERE
sort(table(movies4$genre), decreasing = TRUE)[1:3]
```

```
##
##    Drama   Comedy Thriller
##      402      186      116
```

Your Answer:

Drama, Comedy and Thriller

step b

```r
#WRITE YOUR CODE HERE
movies4$genre2 <- ifelse(movies4$genre == "Thriller" |
                         movies4$genre == "Comedy" |
                         movies4$genre == "Drama",
                         movies4$genre, "Other")

table(movies4$genre2)
```

```
##
##   Comedy    Drama    Other Thriller
##      186      402      101      116
```

Your Answer:

Write your formulated response here.

step c

```r
#WRITE YOUR CODE HERE
model_c <- lm(log_revenue ~ log_budget + genre2 + release_year,
            data = movies4)

summary(model_c)
```

```
##
## Call:
## lm(formula = log_revenue ~ log_budget + genre2 + release_year,
##     data = movies4)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.5777  -0.6653   0.2248   0.8514   3.8557
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     9.097982  19.628713   0.464  0.64320
## log_budget      1.001004   0.047074  21.264  < 2e-16 ***
## genre2Drama    -0.472877   0.162469  -2.911  0.00376 **
## genre2Other    -0.014269   0.223589  -0.064  0.94914
## genre2Thriller -0.321644   0.200858  -1.601  0.10991
## release_year   -0.004184   0.009789  -0.427  0.66926
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.486 on 513 degrees of freedom
##   (289 observations deleted due to missingness)
## Multiple R-squared:  0.5053, Adjusted R-squared:  0.5005
## F-statistic: 104.8 on 5 and 513 DF,  p-value: < 2.2e-16
```

Your Answer:

Write your formulated response here. If the budget increases with 1% than the revenue will increase with 1.001%. The intercept is based on action. If the movie is a drama, Thriller or a different genre in that case

the revenue will decrease compared with an action movie. A higher release year also is associated with a lower revenue. *step d*

```
#WRITE YOUR CODE HERE
model <- lm(revenue ~ release_year + budget + runtime + genre, data = movies4)

new_movie <- data.frame(
  release_year = 2026,
  budget = 20000000,
  runtime = 120,
  genre = "Drama"
)

predict(model, newdata = new_movie, interval = "prediction", level = 0.90)
```

```
##        fit        lwr       upr
## 1 57977522 -156927128 272882171
```

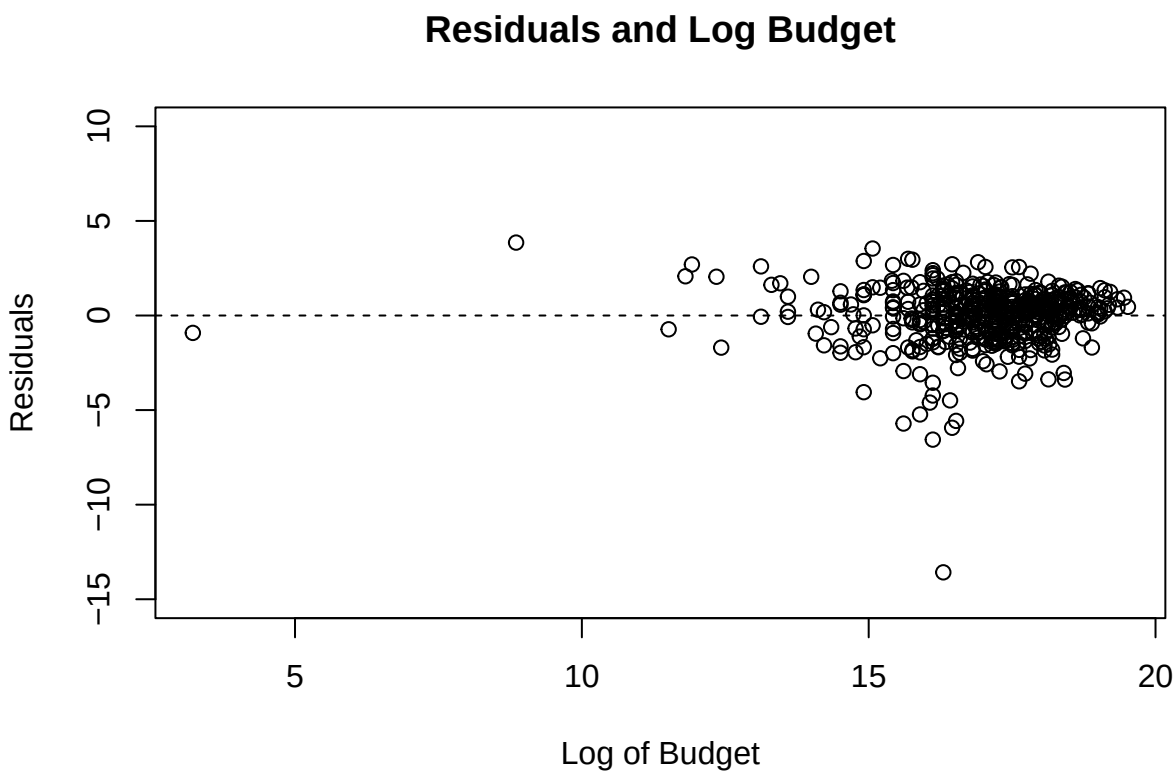
Your Answer:

the predicted revenue is 57 977 522. Almost 58 million of revenue.

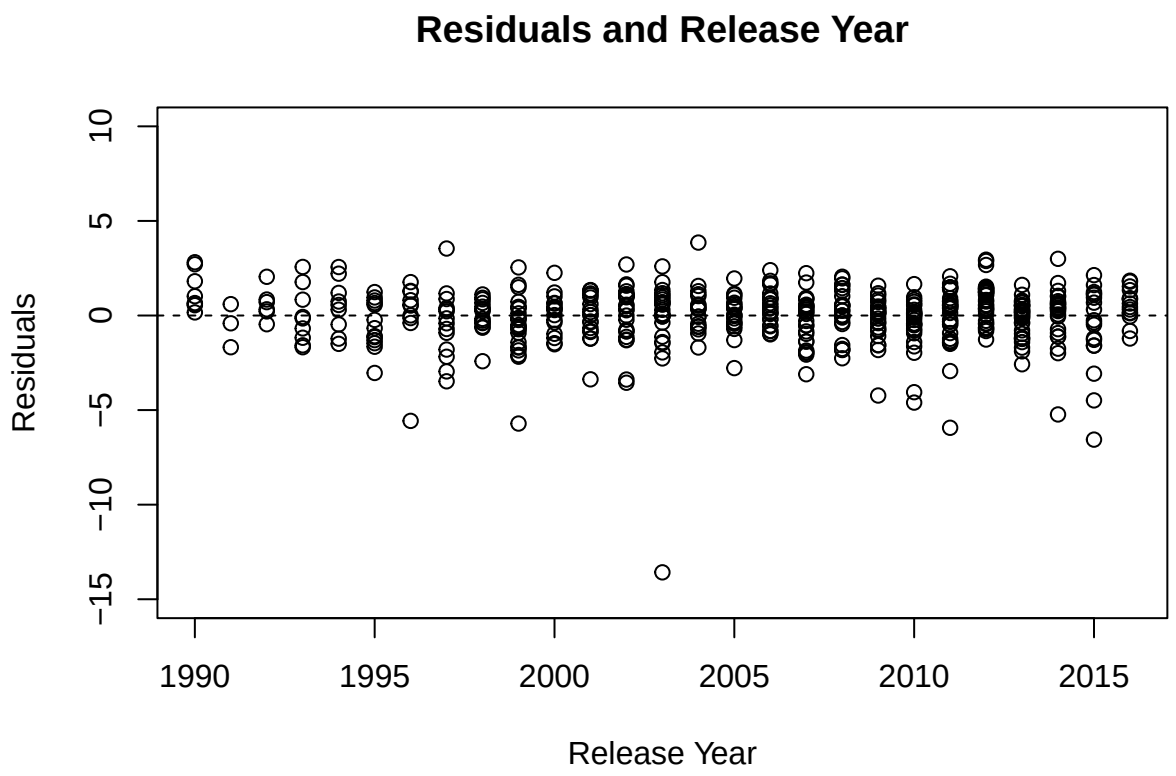
step e

```
#WRITE YOUR CODE HERE
movies_model_c <- model.frame(model_c)
movies_model_c$residuals <- residuals(model_c)

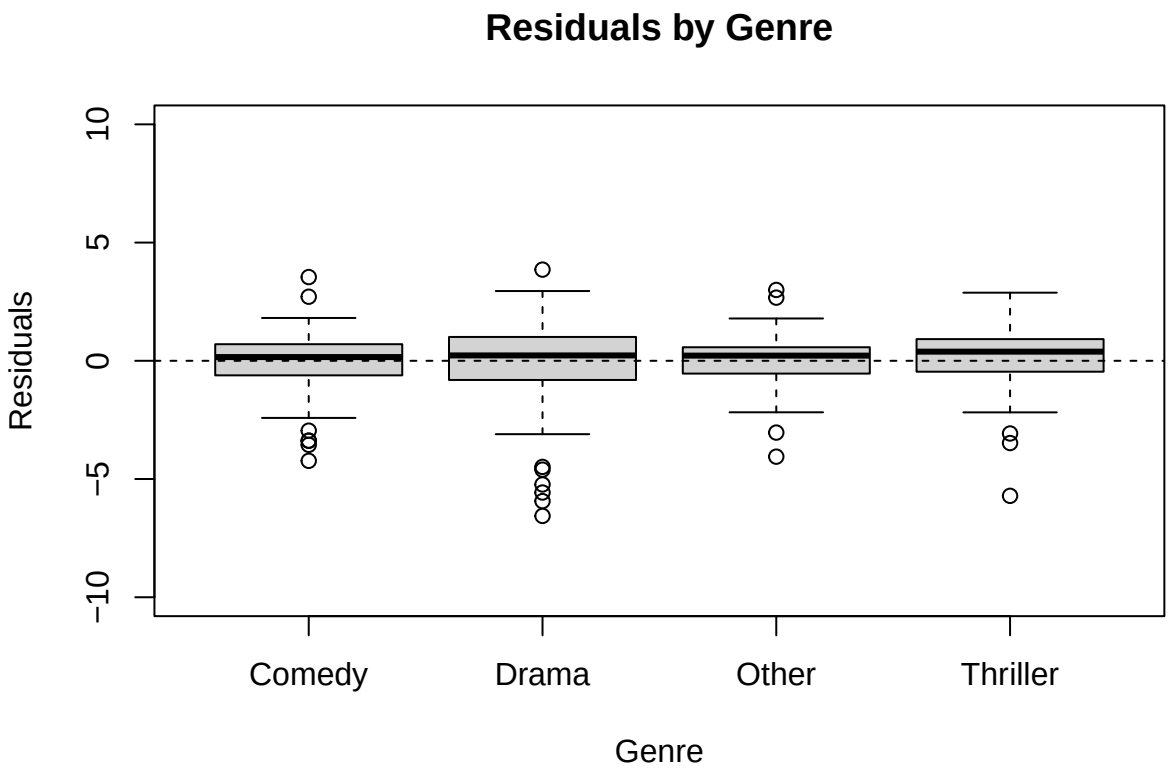
plot(movies_model_c$log_budget, movies_model_c$residuals,
     xlab = "Log of Budget",
     ylab = "Residuals",
     main = "Residuals and Log Budget",
     ylim = c(-15, 10))
abline(h = 0, lty = 2)
```




```
plot(movies_model_c$release_year, movies_model_c$residuals,
     xlab = "Release Year",
     ylab = "Residuals",
     main = "Residuals and Release Year",
     ylim = c(-15, 10))
abline(h = 0, lty = 2)
```




```
boxplot(movies_model_c$residuals ~ movies_model_c$genre2,
        xlab = "Genre",
        ylab = "Residuals",
        main = "Residuals by Genre",
        ylim = c(-10, 10))
abline(h = 0, lty = 2)
```



Your Answer:

The graph about Residuals and Log of Budget shows a bit of heteroskedasticity if you look from the right side of the graph you can see a centralized point which spreads out more the further to the left you go. The other graphs show no sign of heteroskedasticity.

step f

```
#WRITE YOUR CODE HERE
genre2 <- movies4$genre2
movies4$release_month <- as.factor(movies4$release_month)

model_f <- lm(log(revenue) ~ log(budget) + genre2 +
                release_year + release_month,
              data = movies4,
              subset = revenue > 0 & budget > 0)

summary(model_f)
```

```
##
## Call:
## lm(formula = log(revenue) ~ log(budget) + genre2 + release_year +
##     release_month, data = movies4, subset = revenue > 0 & budget >
##     0)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -13.0159  -0.6500   0.1594   0.8586   3.6389
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)     -1.517441  19.708536  -0.077  0.93866
## log(budget)      0.973995   0.047783  20.384  < 2e-16 ***
## genre2Drama     -0.470213   0.162685  -2.890  0.00402 **
## genre2Other     -0.068120   0.226351  -0.301  0.76358
## genre2Thriller  -0.247123   0.200463  -1.233  0.21824
## release_year     0.001324   0.009831   0.135  0.89295
## release_month2  -0.309472   0.370831  -0.835  0.40438
## release_month3  -0.087823   0.365075  -0.241  0.80999
## release_month4  -0.024217   0.369085  -0.066  0.94771
## release_month5   0.369480   0.363217   1.017  0.30953
## release_month6   0.260837   0.364242   0.716  0.47426
## release_month7   0.200592   0.369532   0.543  0.58749
## release_month8  -0.073336   0.354550  -0.207  0.83622
## release_month9  -0.611815   0.334731  -1.828  0.06818 .
## release_month10  0.142468   0.356938   0.399  0.68996
## release_month11  0.440093   0.393189   1.119  0.26355
## release_month12  0.311813   0.350144   0.891  0.37361
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.469 on 502 degrees of freedom
##   (90 observations deleted due to missingness)
## Multiple R-squared:  0.5271, Adjusted R-squared:  0.5121
## F-statistic: 34.98 on 16 and 502 DF,  p-value: < 2.2e-16
```

```
model_f2 <- lm(log(revenue) ~ log(budget) +
                           genre2 + release_year,
                         data = movies4,
                         subset = revenue > 0 & budget > 0)

anova(model_f2, model_f)
```

```
## Analysis of Variance Table
##
## Model 1: log(revenue) ~ log(budget) + genre2 + release_year
## Model 2: log(revenue) ~ log(budget) + genre2 + release_year + release_month
##   Res.Df    RSS Df Sum of Sq     F  Pr(>F)
## 1    513 1133.1
## 2    502 1083.1 11    49.962 2.105 0.01862 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Your Answer: Our H0 is that month of release does not affect revenue and H1 that in at least 1 month, the month has affect on revenue Since the p value in the test is 0.01862 and thus less than 0.05 we reject the null hypothesis, this suggests that the release month has effect has an effect on movie revenue. Therefore we would advice the movie studio that they should consider the release month when releasing a movie.